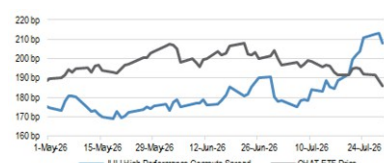


AI Capex Funding: Data Center - Hyperscaler Bond Matrix 2.0

Recent Widening More About Price Than Capacity

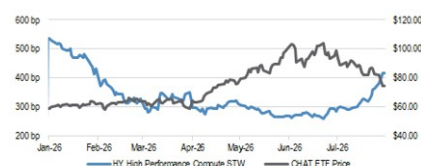
It's been a *mensis horribilis* in AI-related debt. The sell-off in AI equities has received more press, but AI-related debt indices have substantially lagged too. Our High Grade High Performance Compute index has widened 38bp since the start of June to 208bp. For reference, the JPM HY BB Index trades at 201bp. Our High Yield High Performance Compute index has widened 153bp in the last month to 418bp. Data Center AAACMBS has been more resilient, only 5-10bp wider in that same time frame. Some of that speaks to the technical, with substantial AI and data center issuances into the corporate credit market while structured markets have been less busy.

Figure 1: HG HPC Bonds Are Setting New Widens



Source: J.P. Morgan estimates, Bloomberg Finance L.P.

Figure 2: HY HPC Bonds Have Retraced To Feb Levels



Source: J.P. Morgan estimates, Bloomberg Finance L.P.

Data centers have lagged hyperscalers. The underlying corporate relationships between hyperscalers and related project finance bonds have moved since the last version of this report in May. Currently, the average HG project now trades 99bp wide to its related hyperscaler, while the average HY project now trades 208bp wide to its related hyperscaler. The current universe encompasses 23 pairs, up from 15 in May. Interestingly, in May the average HG project finance bond traded 105bp wide to its related hyperscaler - meaning that relationship has tightened (although that might be more a comment on recent rate moves than anything else). Conversely, the average HY project traded 183bp wide to its related hyperscaler in May - so that relationship has widened a meaningful 25bp. We don't think that's a function necessarily of crowding out by wider trading hyperscaler bonds, although at a certain spread level that clearly would happen. Rather, we think that the dynamic is more representative of a relatively orderly repricing of project finance bonds in secondary markets as investors have become more aggressive in dictating pricing and terms in primary markets.

Figure 3: Hyperscaler - Project Bond Relationship Summary

	Count	Average	Wide	Tight
Hyperscaler - HG Project	6	99 bp	179 bp	46 bp
Hyperscaler - HY Project	12	208 bp	305 bp	117 bp
HY Lessee - HY Project	5	-252 bp	100 bp	-407 bp

Source: J.P. Morgan.

Recent bond performance has been less about market capacity and more about investor ability to dictate. For context, pre and immediately post GFC,

See page 10 for analyst certification and important disclosures.

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large USD HG issuers routinely represented 3–8% of the benchmark for extended periods of time. We are clearly nowhere near these numbers yet either individually or collectively for the HPC cohort. By our estimates, the hyperscalers could collectively issue ~\$1.7tr of additional USD HG debt before they would hit a theoretical 3% index-weight, a level which could trigger more risk-limit scrutiny. Note we are including related data center debt for which the hyperscaler is a lessee in our number as we believe most investors are managing risk on that basis. See [Growing Pains](#) for a much more detailed discussion of that dynamic.

Despite the growing pains, the universe keeps getting bigger. The combined hyperscaler and data center universe across the high grade and high yield indices encompasses 31 different issuers with over \$576 billion in outstanding bonds as well as over \$5 billion and growing of leveraged loans. We expect the universe to continue to grow, and therefore we expect this to continue to be a living document, and we welcome any and all feedback. New to this edition we have also included thoughts around structured products, including 3 CMBS and 2 ABS deals accounting for over \$4 billion of investable instruments.

Figure 4: Hyperscaler & AI/Data Center Investable Universe (\$bn)

	Direct IG Index Debt	Proportionate Share of DC Lessor Debt		Total Related Obligations
		IG Index	HY Index	
AMZN	\$121	\$0	\$3	\$124
MSFT	\$34	\$5	\$1	\$40
GOOGL	\$51	\$3	\$13	\$67
META	\$84	\$40	\$7	\$131
ORCL	\$117	\$14	\$3	\$134
Other	\$73	\$4	\$29	\$106
Total	\$480	\$66	\$56	\$602

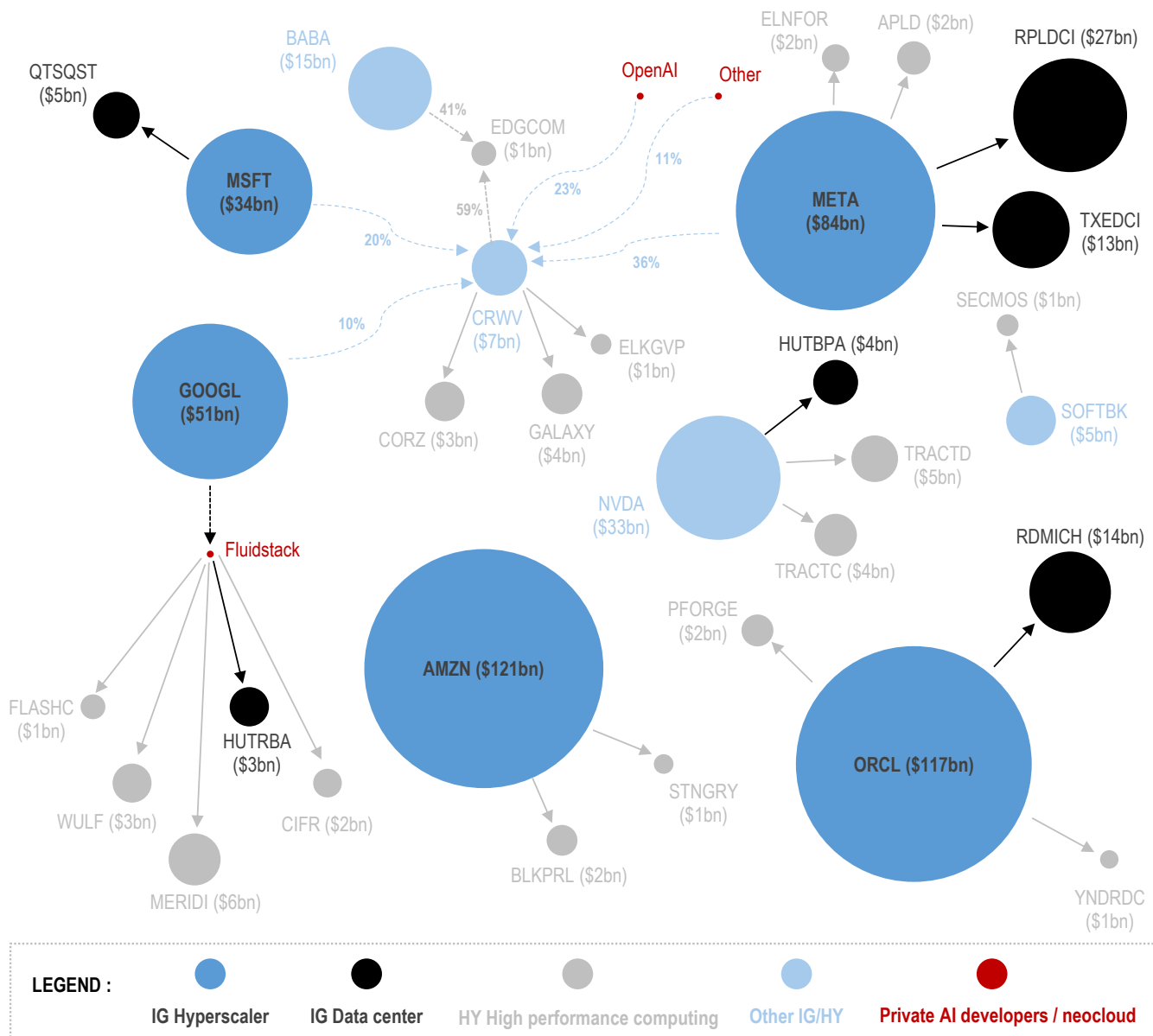
Source: J.P. Morgan.

AI/Hyperscaler Ecosystem Continues To Grow

We are reaching the proverbial limits of the increasingly inaccurately named credit linkages simplified chart. However, we do think it's important to clarify just how central the major hyperscalers are to the overall ecosystem even as it grows and evolves.

Figure 5: Hyperscalers and the HPC Universe Credit Linkages Simplified

Note: Solid arrows: single-tenant projects. Dotted arrows: multi-tenant or intermediary structures; % reflects estimated revenue share



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 6: IG and HY High Performance Computing versus Hyperscaler Bond Relative Value

Note: The blue rows are IG HPC bonds. The gray rows are HY HPCs which are compared on both YTM and YTC basis

Security	Issuer	Par amount outstanding \$bn	Rating (Mdy / SP / Fitch)	Current G- Spread (bp)	Rating notch difference / Basis, bp
RDMICH 7 1/2 03/30/45	RD Michigan	14.0	Baa3 / - / BBB	316	1 notches
ORCL 5 3/8 07/15/40	Oracle Corp	2.2	Baa2 / BBB- / BBB	247	69
RPLDCI 6.581 05/30/49	Beignet Investor LLC	27.3	- / A+ / -	203	1 notches
META 6.2 05/15/46	Meta Platforms Inc	4.0	Aa3 / AA- / -	157	46
TXEDCI 7.534 11/30/48	Sopaipilla Investor LLC	12.6	- / A+ / AA-	225	1 notches
META 6.2 05/15/46	Meta Platforms Inc	4.0	Aa3 / AA- / -	157	68
QTSQST 5.7 04/15/36	QTS Fayetteville	4.6	Baa2 / - / -	234	8 notches
MSFT 3.45 08/08/36	Microsoft Corp	1.7	Aaa / AAA / -	56	179
HUTRBA 6.192 11/15/42	Hut 8 DC LLC	3.3	- / BBB- / BBB-	193	7 notches
GOOGL 4.8 02/15/36	Alphabet Inc	4.3	Aa2 / AA+ / -	76	117
HUTBPA 6.129 11/30/42	Beacon Point DC LLC	4.3	Baa2 / - / -	196	5 notches
NVDA 3 1/2 04/01/40	Nvidia Corp	1.0	Aa1 / AA / WD	84	112
BLKPRL 6 1/8 02/15/31	Black Pearl Compute	2.0	Ba2 / - / BB-	177	9 notches
AMZN 4 1/4 03/13/31	Amazon.com Inc	5.0	A1 / AA / AA-	60	117
AMZN 3.85 03/13/28	Amazon.com Inc	2.3	A1 / AA / AA-	26	152
PFORGE 6 3/4 03/15/31	APLD ComputeCo 2 LLC	2.2	Ba3 / - / BB-	308	4 notches
ORCL 4.95 02/04/31	Oracle Corp	3.5	Baa2 / BBB- / BBB	174	134
ORCL 2.3 03/25/28	Oracle Corp	2.0	Baa2 / BBB- / BBB	111	197
YNDRDC 6 7/8 06/30/31	Yondr JK 1 LLC	0.7	Ba3 / - / BB	394	4 notches
ORCL 2 7/8 03/25/31	Oracle Corp	3.3	Baa2 / BBB- / BBB	174	220
ORCL 4 1/2 05/06/28	Oracle Corp	0.8	Baa2 / BBB- / BBB	105	289
FLASHC 7 1/4 12/31/30	Flash Compute LLC	1.3	Ba3 / - / -	267	10 notches
GOOGL 4.1 11/15/30	Alphabet Inc	2.5	Aa2 / AA+ / -	48	219
GOOGL 0.8 08/15/27	Alphabet Inc	1.0	Aa2 / AA+ / -	11	256
WULF 7 3/4 10/15/30	Wulf Compute LLC	3.2	Ba2 / BB / BB	206	9 notches
GOOGL 4.1 11/15/30	Alphabet Inc	2.5	Aa2 / AA+ / -	48	158
GOOGL 0.8 08/15/27	Alphabet Inc	1.0	Aa2 / AA+ / -	11	195
MERIDI 6 1/4 04/30/31	Meridian Arc Holdco LLC	5.7	Ba2 / BB- / BB	262	9 notches
GOOGL 4.1 02/15/31	Alphabet Inc	3.0	Aa2 / AA+ / -	46	216
GOOGL 3 7/8 11/15/28	Alphabet Inc	1.0	Aa2 / AA+ / -	19	243
CIFR 7 1/8 11/15/30	Cipher Compute LLC	1.7	Ba3 / - / BB-	217	10 notches
GOOGL 4.1 11/15/30	Alphabet Inc	2.5	Aa2 / AA+ / -	48	169
GOOGL 0.8 08/15/27	Alphabet Inc	1.0	Aa2 / AA+ / -	11	205
TRACTC 5 7/8 03/01/31	SV RNO Property Owner 1	3.8	Ba1 / BB+ / BB	317	7 notches
NVDA 2 06/15/31	Nvidia Corp	1.3	Aa1 / AA / -	41	275
NVDA 1.55 06/15/28	Nvidia Corp	1.3	Aa1 / AA / -	21	295
TRACTD 6 1/2 05/01/31	PR RNO Property Owner 1	4.6	Ba1 / BB / BB	346	7 notches
NVDA 2 06/15/31	Nvidia Corp	1.3	Aa1 / AA / -	41	305
NVDA 1.55 06/15/28	Nvidia Corp	1.3	Aa1 / AA / -	21	325
APLD 9 1/4 12/15/30	APLD ComputeCo LLC	2.4	- / BB- / BB-	306	9 notches
META 4.2 11/15/30	Meta Platforms Inc	4.0	Aa3 / AA- / -	59	247
META 4.6 05/15/28	Meta Platforms Inc	1.5	Aa3 / AA- / -	27	279
CORZ 7 3/4 05/15/31	Core Scientific Finance	3.3	- / BB / BB-	396	-2 notches
CRWV 9 02/01/31	Coreweave Inc	1.8	B1 / B / BB-	766	-370
SECMOS 8 7/8 05/01/31	SE Cosmos LLC	1.0	- / BB+ / BB-	511	0 notches
SOFTBK 8 1/4 10/22/31	Softbank Group Corp	0.6	- / BB+ / -	410	100
SOFTBK 5 04/15/28	Softbank Group Corp	1.1	Ba2u / BB+ / -	199	311
EDGCOM 7 1/2 04/30/31	Edged Compute LLC	1.3	- / BB+ / BB-	454	-3 notches
CRWV 9 02/01/31	Coreweave Inc	1.8	B1 / B / BB-	766	-312
ELKGVP 7 1/2 06/15/31	Elk Grove Village Proper	0.9	Ba3 / BB- / BB-	415	-1 notches
CRWV 9 3/4 10/01/31	Coreweave Inc	2.8	B1 / B / BB-	823	-407
GALAXY 9 7/8 08/01/31	Galaxy Helios Data Cntr	3.5	- / BB- / BB-	552	-1 notches
CRWV 9 3/4 10/01/31	Coreweave Inc	2.8	B1 / B / BB-	823	-271
STNGRY 6 06/15/31	Stingray Compute	0.8	Ba2 / - / BB-	224	9 notches
AMZN 3.7 06/12/31	Amazon.Com Inc	2.5	A1 / AA / AA-	64	160
AMZN 1.65 05/12/28	Amazon.Com Inc	2.3	A1 / AA / AA-	25	199
ELNFOR 7 06/15/31	APLD ComputeCo 3 LLC	1.6	- / BB- / BB-	347	9 notches
META 4.55 05/15/31	Meta Platforms Inc	3.0	Aa3 / AA- / -	73	274
META 4.6 05/15/28	Meta Platforms Inc	1.5	Aa3 / AA- / -	27	320

Source: J.P. Morgan, Bloomberg Finance L.P. Data as of 07/30/2026

High Grade market performance is more about price than capacity

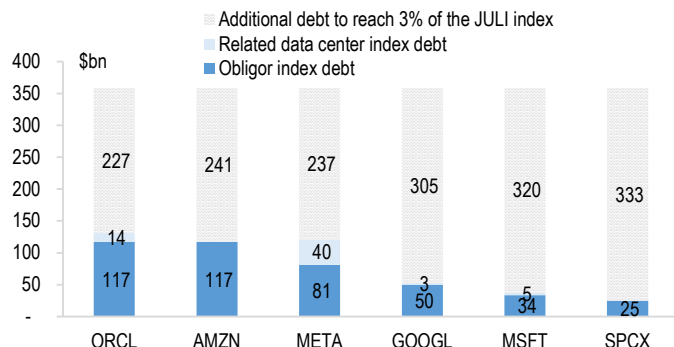
History and global markets suggest capacity for hyperscaler issuance remains substantial. Concerns that rapidly growing AI-related issuance is pushing insurers toward single-issuer concentration limits are overstated. In fact, the USD HG market remains far less concentrated than in prior cycles. Pre and immediately post GFC, large USD HG issuers routinely represented 3–8% of the benchmark for extended periods of time, while today smaller and less liquid non-USD corporate credit markets currently function well with issuers accounting for 5–17% of their respective indices. By our estimates, the hyperscalers could collectively issue ~\$1.7tr of additional USD HG debt before they would hit a theoretical 3% index-weight, a level which could trigger more risk-limit scrutiny.

Looks can be deceiving. Insurers' assess issuer limits against their broader pool of invested assets rather than solely against public HG holdings. As a result, issuers that appear large in the HG index translate to materially smaller weights in insurer portfolios, and life insurers entered 2026 underweight hyperscaler HG debt. Insurers invested assets are growing rapidly on account of strong annuity sales and the allocation to public HG is higher too, which bodes well for future demand.

The market-wide spread impact of hyperscaler releveraging is de-minimis. We calculate inside a custom 'hyperscaler index' spread. Despite significant issuance and material spread widening of hyperscaler debt, the rest of the market is well contained so far and thus we estimate hyperscaler widening is only worth 3bp of widening for the broader market YoY. But Technology has just surpassed US Banks as the largest sector in JULI at 11.6% of the index and has accounted for 33% of net HG issuance YTD so the marketwide impact of hyperscaler spread moves will continue to grow.

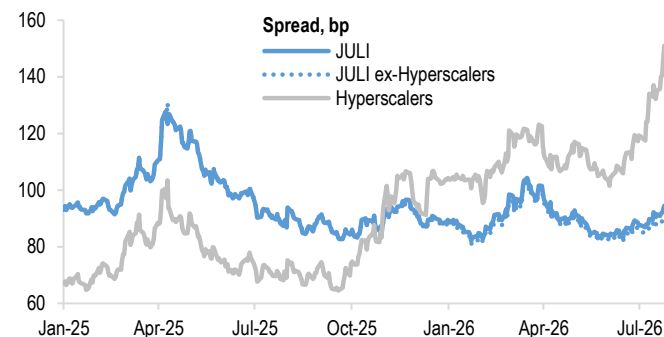
The current hyperscaler widening is a byproduct of the HG investor community trying to rationally price in an accelerating pace of issuance from the hyperscalers and not a reflection of insurers already hitting issuer risk limits for these issuers. Thus, demand for hyperscaler and associated datacenter financings should remain robust, albeit at a price that will continue to be very dynamic as expectations for future funding needs and subsequent monetizations efforts become clearer. Put another way, investors are beginning to dictate price and terms more aggressively as the issuance wave has continued. We discuss all this in much more detail in our recent [Growing Pains](#) report.

Figure 7: Estimating \$1.7tr of USD HG hyperscaler debt capacity for each to hit 3% of the USD HG index



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 8: The impact of hyperscaler spread widening has been de-minimis on the broader IG index



Source: J.P. Morgan

New issues priced well wide of the index tend to perform well but the 2026 cohort is the exception thus far

Another large datacenter financing priced this week in HG for a total of \$12.3bn, the second largest tranche YTD after RDMICH. Notable to us was the very wide pricing spread of 287.5bp over UST 10yr; this is the widest A or higher rated new issue in the past 3 years. Moreover, it is the widest A+ data center financing since the Beignet 30yr bond priced at +225bp in October 2025. We think there are two interesting takeaways from a strategy perspective:

The spread and duration of the new datacenter security imply an approximate excess return breakeven of ~31bp which is 3.6x higher than at the index level currently for the 30yr part of the curve and 2.2x for the 10yr part of the curve. Bonds that price well wide of the index have tended to perform well in subsequent months, but this hasn't been the case so far in 2026 (we wrote about this from a different angle recently here). To delve into this, we ran an analysis since 2020 of newly issued bonds with issuance spreads of 2x or wider to that of the JULI spread at the same point in time. The first notable result is that more bonds fit into this category in 2026 than any past year in our analysis, including 2020. The vast majority of this issuance YTD has come from hyperscalers, datacenters and BDCs.

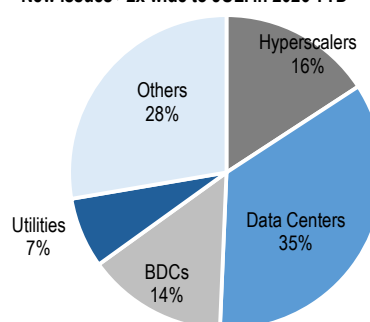
Figure 9: Record share of new issues pricing >2x wide to JULI in 2026 YTD

Year	\$bn of bonds pricing at 2x JULI spread	Total Issuance (\$bn)	% share
2020	77	1,758	4.4%
2021	14	1,381	1.0%
2022	33	1,214	2.7%
2023	43	1,207	3.5%
2024	56	1,511	3.7%
2025	85	1,639	5.2%
2026	85	1,325	6.4%

Source: J.P. Morgan

Figure 10: Very wide new issues mostly tied to AI Capex YTD

New issues >2x wide to JULI in 2026 YTD



Source: J.P. Morgan

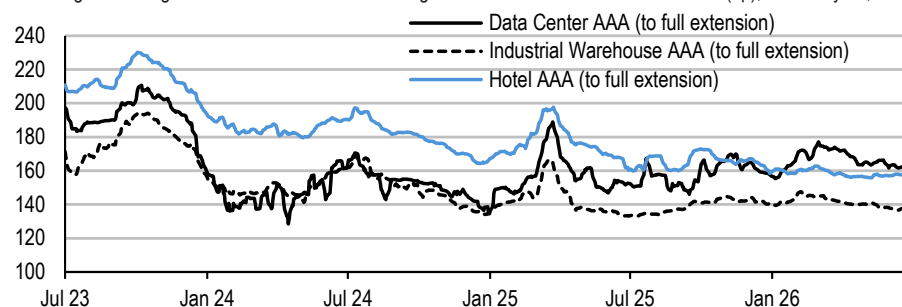
In terms of the performance of this cohort post issuance, on average in 2022-2025 (the ‘high yield’ regime years) these wide new issue deals tightened 22bp from issuance spread in the first month and 26bp by the end of their 3rd month, with the long end cohort tending to do better (30yr bucket averaged 42bp tighter at 1m). The pattern was consistent rather than driven by a handful of names, and it held across the curve. However, this pattern has so far broken in 2026: the 2026 cohort of new issues 2x wide to JULI were roughly 10bp wider one month after pricing and 21bp wider at three months, a clear break from prior years, and the long end has been the weakest point on the curve in 2026, 22bp wider at 1m and 25bp wider at 3m (for the bonds where there is sufficient historical data to derive the spread change). This new trend is clearly weighing on the broader market and so we are watching closely to see if the current cohort of wider new issues begins to effect a shift back towards the historical pattern or not.

Securitized products markets have been more resilient

Securitized data centers have so far held up reasonably well, with SASB data center AAAs widening by about 5bp week-over-week, but underperforming core SASB sectors like industrial warehouse and hotel AAAs that were about unchanged week-over-week (Figure 11). Securitized data centers are also more diversified in terms of tenancy and use cases than the kind of data centers that are being financed in the corporate bond market. While a growing chunk of securitized data centers are leased to hyperscalers, a significant amount of outstanding securitized data centers (ABS in particular) have exposure to enterprise/wholesale and retail colocation tenants. These tenants and their data center use cases are less impacted by the recent AI capex-related angst the corporate and equity markets are facing. Importantly, the securitized data center bonds are not very liquid, so they could be lagging the far larger corporate bond markets.

Figure 11: CMBS data center AAAs widened modestly this past week, underperforming core CMBS sectors

J.P. Morgan Floating-Rate SASB Index discount margin to 1m Term SOFR to full extension (bp), as of July 29, 2026



Source: J.P. Morgan, Pricing Direct

Unlike the corporate bond market, CMBS and ABS data center tenant identities are typically confidential, which makes a similar data center versus unsecured bond relative value analysis difficult. Instead, the relative value analysis within securitized data centers focuses more on tenant type and structure. Across our select CMBS and ABS data center bond comparisons, CMBS generally comes out looking wider, though structural differences and tenant lease length and durability should be considered (Figure 12).

Figure 12: CMBS data center bonds generally look wider than ABS data center bonds, though structural differences and tenant lease length and durability should be considered

Indicative spreads for select CMBS and ABS data center bonds (bp), as of July 29, 2026

Product	Data Center/Tenant Type	Security	Issuer	Par amount outstanding (\$bn)	Rating (Mdy / SP / Fitch / DBRS or KBRA)	Current Spread or DM	Rate Format
CMBS	Hyperscale	VDCM 2025-AZ A	Retained Vantage Data Centers, LP	0.36	- / - / AAA / AAA	135 / 160	Fixed-rate (to Treasuries / to SOFR swaps)
CMBS	Hyperscale	VDCM 2025-AZ C	Retained Vantage Data Centers, LP	0.11	- / - / - / AL	215 / 240	Fixed-rate (to Treasuries / to SOFR swaps)
CMBS	Hyperscale	BX 2025-VLT7 A	Blackstone, QTS	0.75	- / - / AAA / AAA	173	Floating-rate (to 1m Term SOFR)
CMBS	Hyperscale	BX 2025-VLT7 C	Blackstone, QTS	0.10	- / - / A- / AA	240	Floating-rate (to 1m Term SOFR)
ABS	Hyperscale	QTSII 2026-7 A-21	QTS Issuer ABS II LLC	0.23	- / - / AAA / - / -	137 / 174	Fixed-rate (to Treasuries / to SOFR swaps)
ABS	Hyperscale	QTSII 2026-7 A-23	QTS Issuer ABS II LLC	0.06	- / A- / A / - / -	189 / 226	Fixed-rate (to Treasuries / to SOFR swaps)
CMBS	Colocation	SWCH 2025-DATA A	Switch	1.16	- / - / AAA / AAA	175	Floating-rate (to 1m Term SOFR)
CMBS	Colocation	SWCH 2025-DATA C	Switch	0.15	- / - / A- / AAL	245	Floating-rate (to 1m Term SOFR)
ABS	Colocation	SWTCH 2026-1 A-21	Switch ABS Issuer, LLC	0.54	- / - / - / AAA / -	138 / 165	Fixed-rate (to Treasuries / to SOFR swaps)
ABS	Colocation	SWTCH 2026-1 A-23	Switch ABS Issuer, LLC	0.11	- / - / - / AL / -	207 / 234	Fixed-rate (to Treasuries / to SOFR swaps)

Note: ABS highlighted in gray. ABS spreads are estimated using Pricing Direct bid prices run to ARD.

Source: J.P. Morgan, Pricing Direct

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**Percentage of subject companies within each of the "Overweight," "Neutral" and "Underweight" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

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